

Jerome Banks

Oakland, CA | jerome_banks@yahoo.com | linkedin.com/in/jerome-banks-8320771 | github.com/jeromebanks

Summary

Staff-level data platform engineer with 15+ years in data platforms, building distributed data infrastructure from Hadoop/Hive and Spark through GCP scientific workflows. Founded the open-source Brickhouse (Hive UDFs) and Satisfaction (asset-based Hadoop scheduler) projects, later built Cubism for multidimensional Spark aggregation, and has repeatedly been brought in as the senior IC to stand up big data capability from scratch. Most recently at GSK, migrated imaging and genomics pipelines to Nextflow/GCP Batch and built AI-assisted pipeline tooling for the platform team.

Skills

Big Data & Distributed Systems: Apache Spark, Hadoop/MapReduce/HDFS, Apache Beam/Dataflow, Apache Hive, Delta Lake, Databricks, Apache Kafka, Akka

Cloud Platforms: Google Cloud Platform (BigQuery, Dataflow, Dataproc, Batch, Pub/Sub), AWS (EMR, S3, Kinesis, DynamoDB, Lambda)

Languages: Scala, Python, Java, Ruby, C++

Orchestration & DevOps: Nextflow/DSL2, Apache Airflow, Kubernetes/Helm, Docker, Terraform, Prometheus/Grafana, pytest/WireMock

AI Engineering: Model Context Protocol (MCP) / FastMCP, Claude Code Skills, Gemini API, GPU pipelines (PyTorch; A100/T4/L4)

Experience

GSK — Senior Data Platform Engineer

07/2023 – 06/2026

- Migrated a phenomics image-processing pipeline from Vertex AI to Nextflow DSL2 on GCP Batch, driving GPU utilization from ~1.75% to ~100% and cutting single-barcode runtime from ~2h11m to ~43m32s.
- Built nf-forge, an AI-assisted approach for generating Nextflow pipelines from legacy scientific code, first proven by migrating an HRD-determination workflow to production in ~4–5 days.
- Built a ~12-tool FastMCP server exposing GCP Batch/Cloud Run/Workflows/GCS for pipeline debugging, adopted by several DPE teammates.
- Designed a distributed 3D cell-segmentation pipeline (Nextflow, Cellpose-SAM on L4 GPUs) for multi-hundred-GB microscopy volumes, using streamed TIFF-to-Zarr conversion and halo-overlapped block partitioning, and fixed a lost-update bug from concurrent Zarr chunk writes; production-ready ~10 weeks after first exposure.
- Ingested 150TB+ of the Broad Institute's JUMP-CP Cell Painting images (millions of files) into GCP via Dataflow.
- Took GCP Workflow development from zero test coverage to per-workflow pytest integration tests, WireMock-based error simulation, an end-to-end test harness, and multi-environment CI/CD across 12+ workflows.

Apixio — Staff Software Engineer, Data Platform

08/2022 – 07/2023

- Eliminated chronic daily failures in a Scala/Spark healthcare document-processing pipeline by buffering per-record Delta Lake telemetry writes through Kafka into batched updates.
- Built a Spark zombie-process detector that caught silent stalls before they missed time-critical, revenue-impacting deadlines.
- Introduced Prometheus/Grafana observability where none existed.

Tatari — Staff Data Platform Engineer

02/2021 – 07/2022

- Brought in as the most senior IC to introduce Big Data to a TV ad-tech company with no Spark experience or budget for managed services; stood up production Spark on Kubernetes (AWS EKS) as a zero-cost on-ramp.
- Built a multi-dimensional OLAP-style Spark aggregation tool for combined linear and streaming TV campaign metrics; shipped a Scala Spark UDF/UDAF library for team adoption.

Demandbase — Principal Big Data Engineer

09/2018 – 02/2021

- Owned the core data pipelines behind Demandbase's intent-signal product from conception through production support.
- Re-architected a multi-dimensional web-engagement aggregation pipeline (Cubism pattern), adding geo while making it substantially faster and cheaper.
- Reduced BigQuery slot usage by roughly an order of magnitude on legacy pipelines that had been failing SLAs.

Jumpshot/Avast — Senior Software Engineer

12/2016 – 08/2018

- De facto technical lead for a data processing team working with an opt-in clickstream panel of ~100M devices and 435M monthly active users.
- Built Cubism, a multidimensional Spark aggregation library adopted by both the San Francisco and Czech Republic teams; basis for the FunnelCube e-commerce insights product.
- Extended Cubism to Spark Streaming with Kafka-based data-quality monitoring, catching deployment misconfigurations within minutes.

StitchFix — Data Platform Engineer, Algorithms & Analytics

08/2015 – 10/2016

- Built production web-event ingestion (HTTP/AMQP into AWS S3 via Kinesis) using Akka Reactive Streams.
- Built a Spark DataSource connector for distributed read/write of ML model weights in DynamoDB, used in production by the algorithms team.

Tagged.com / if(we) — Senior Software Engineer, Analytics Infrastructure

03/2014 – 07/2015

- Completed and shipped Satisfaction (Hadoop batch scheduler) to production, running event ingestion for a platform with 300M registered members and demographic aggregates across millions of segments.

Klout — Senior Software Engineer

09/2011 – 02/2014

- Built the Hadoop/Hive pipelines behind Klout's topical influence scores — 10,000+ topics scored daily across hundreds of millions of users — contributing to two published papers and a co-invented patent filing.
- Founded and open-sourced Brickhouse, a Hive UDF library that let a single SQL query express pipeline patterns otherwise written as multi-stage MapReduce jobs.

Quantcast — Senior Software Engineer

09/2008 – 09/2011

- Developed and maintained the Publisher daily reach job — Quantcast's flagship measurement product, which ran in production for over a decade.

Earlier (pre-2008): Cyanea Systems (acquired by IBM) — built bytecode-instrumentation data collection for a J2EE APM product resold as IBM WebSphere Studio Application Monitor; also enterprise Java and financial trading systems.

Selected Projects

Brickhouse (github.com/klout/brickhouse): 573 forks and 33 external contributors on GitHub; published to Maven Central. Its declarative aggregation ideas were later re-implemented in Spark as Cubism.

Satisfaction (github.com/ifwe/satisfaction): Asset-based Hadoop batch scheduler in Scala (DSL, Play! UI, Akka actor dependency engine) that tracks which data assets are satisfied and rebuilds only unsatisfied dependencies on failure; taken from prototype at Klout to production at if(we) as an alternative to Oozie/Jenkins.

Talent Brain (github.com/jeromebanks/talent-brain): Open-source (MIT) agentic career-profile system: a markdown knowledge graph plus Claude Code skills for ingesting, deepening, and generating career documents; used to maintain this profile.

Apache Hivemall: committer.

Education

California Institute of Technology — B.S., Engineering & Applied Science (Computer Science emphasis).

Publications & Patents

Acknowledged contributor on two published research papers (LASTA, KDD '14; Klout Topics, K-CAP '17); co-inventor on four granted US patents in one APM patent family (Cyanea Systems/IBM) plus a Klout patent filing.